

Formative Assessment 4: Applied Regression Analysis

Dummy-Variable Regression Using Real-World Data

Dataset: diamonds

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Part A. Exploring and Preparing the Data

The analysis uses the `diamonds` dataset from the `ggplot2` package. The dataset contains 53,940 observations and 10 variables.

The R commands used to examine the dataset are:

```
1 # Load the ggplot2 package
2 library(ggplot2)
3
4 # Load the diamonds dataset
5 data(diamonds)
6
7 # View the first few observations
8 head(diamonds)
9
10 # Check the structure of the dataset
11 str(diamonds)
12
13 # Get summary statistics
14 summary(diamonds)
15
16 # Number of observations
17 nrow(diamonds)
18
19 # Number of variables
20 ncol(diamonds)
```

1. Number of observations and variables

The `diamonds` dataset contains:

53,940 observations and 10 variables

2. Response variable

The response variable is **price**. It measures the price of a diamond in US dollars.

3. Quantitative explanatory variable

The quantitative explanatory variable is **carat**. It measures the weight of the diamond in carats.

4. Categorical explanatory variable

The categorical explanatory variable is **cut**. It describes the quality of the diamond's cut.

The five categories are:

- Fair
- Good
- Very Good
- Premium
- Ideal

5. Reference category

The reference category selected is **Ideal**.

A reference category is necessary because a categorical variable with five categories requires four dummy variables when an intercept is included in the regression model. The coefficients of the dummy variables are then interpreted as differences relative to the reference category.

The following R commands set **Ideal** as the reference category:

```
1 diamonds$cut <- factor(diamonds$cut, ordered = FALSE)
2 diamonds$cut <- relevel(diamonds$cut, ref = "Ideal")
3
4 levels(diamonds$cut)
```

Part B. Constructing Dummy Variables

With `Ideal` as the reference category, four dummy variables are required.

1. Dummy-variable coding table

Table 1: Dummy-variable coding for the cut variable

Cut	Fair	Good	Very Good	Premium
Fair	1	0	0	0
Good	0	1	0	0
Very Good	0	0	1	0
Premium	0	0	0	1
Ideal	0	0	0	0

The R commands used to construct the dummy variables are:

```
1 diamonds$Fair <- ifelse(diamonds$cut == "Fair", 1, 0)
2 diamonds$Good <- ifelse(diamonds$cut == "Good", 1, 0)
3 diamonds$Very_Good <- ifelse(diamonds$cut == "Very Good", 1, 0)
4 diamonds$Premium <- ifelse(diamonds$cut == "Premium", 1, 0)
5
6 head(diamonds[, c("cut", "Fair", "Good",
7                   "Very_Good", "Premium")])
```

2. Meaning of the all-zero category

The all-zero category represents the reference category, which is **Ideal**. Therefore, when Fair, Good, Very Good, and Premium are all equal to zero, the observation belongs to the Ideal cut category.

3. Why five dummy variables should not be included

Five dummy variables should not be included together with an intercept because the five dummy variables always sum to one. This would create perfect multicollinearity, also known as the **dummy-variable trap**. Therefore, with five categories and an intercept, only four dummy variables are included.

Part C. Additive Dummy-Variable Regression

The additive model fitted in R is:

```
1 model1 <- lm(price ~ carat + cut, data = diamonds)
2 summary(model1)
```

The model assumes that the different cut categories have different intercepts but a common slope for carat.

1. Estimated regression equation

Using `Ideal` as the reference category, the estimated regression equation is:

$$\begin{aligned}\widehat{Price} = & -2074.546 + 7871.082(\text{carat}) \\ & - 1800.924(\text{Fair}) - 680.592(\text{Good}) \\ & - 290.789(\text{Very Good}) - 361.847(\text{Premium}).\end{aligned}$$

Here, each dummy variable equals 1 when the diamond belongs to that cut category and 0 otherwise.

2. Interpretation of the carat coefficient

The estimated coefficient of carat is:

$$7871.082$$

Holding cut constant, a one-carat increase in diamond weight is associated with an estimated increase of approximately

$$\$7,871.082$$

in predicted diamond price.

3. Interpretation of one dummy-variable coefficient

Consider the Fair coefficient:

$$-1800.924$$

Holding carat constant, a diamond with a Fair cut has a predicted price approximately \$1,800.924 lower than a diamond with an Ideal cut, on average.

4. Interpretation of R^2

The additive model has:

$$R^2 = 0.856$$

Therefore, approximately

$$85.647\%$$

of the variation in diamond price is explained by carat and cut in the additive model.

5. Same or different slopes?

The regression lines have the **same slope** for the different cut categories in the additive model.

This is because the model contains carat and cut but does not contain carat-by-cut interaction terms. The cut categories therefore change the intercepts, while the estimated effect of carat remains the same.

Part D. Interaction Between Carat and Cut

The interaction model is:

```
1 model2 <- lm(price ~ carat * cut, data = diamonds)
2
3 summary(model2)
```

The interaction model allows the effect of carat on price to vary across the different cut categories.

1. Hypotheses

The null hypothesis is:

H_0 : There is no carat-by-cut interaction.

That is, the effect of carat on price is the same for all cut categories.

The alternative hypothesis is:

H_1 : There is a carat-by-cut interaction.

That is, the effect of carat on price differs across cut categories.

2. Interaction coefficients and p-values

The interaction terms from the regression output are:

Table 2: Interaction coefficients		
Interaction	Estimate	p-value
Carat $\times$ Fair	-2267.896	8.195×10^{-194}
Carat $\times$ Good	-712.755	1.175×10^{-41}
Carat $\times$ Premium	-384.639	2.414×10^{-29}
Carat $\times$ Very Good	-256.419	1.317×10^{-11}

All four interaction terms have p-values below 0.05.

3. Incremental F-test / ANOVA

The appropriate comparison is between the additive model and the interaction model:

```
1 anova(model1, model2)
```

The incremental F-test gives:

$$F = 247.735$$

with a p-value of approximately

$$p = 2.893 \times 10^{-211}.$$

The comparison adds four interaction terms, so the numerator degrees of freedom is

4. The residual degrees of freedom for the interaction model is 53,930.

Table 3: Incremental F-test comparing the additive and interaction models

Model	Res. DF	RSS	Df	F	p-value
Additive	53934	—	—	—	—
Interaction	53930	—	4	247.735	2.893×10^{-211}

4. Statistical decision at $\alpha = 0.05$

Because

$$p < 0.05,$$

we reject the null hypothesis.

There is statistically significant evidence that the effect of carat on diamond price differs among the cut categories.

5. Practical difference between the models

The additive model assumes that the relationship between carat and price has the same slope for every cut category. The interaction model allows the slope relating carat to price to vary depending on the diamond's cut.

Part E. Visualization and Statistical Conclusion

The required scatterplot was created using the following R commands:

```
1 diamond_plot <- ggplot(  
2   diamonds,  
3   aes(x = carat, y = price, color = cut)  
4 ) +  
5   geom_point(alpha = 0.3) +  
6   geom_smooth(  
7     method = "lm",  
8     se = FALSE  
9   ) +  
10  labs(  
11    title = "Diamond Price versus Carat by Cut",  
12    x = "Carat",  
13    y = "Price (US Dollars)",  
14    color = "Cut"  
15  ) +  
16  theme_minimal()  
17  
18 print(diamond_plot)
```

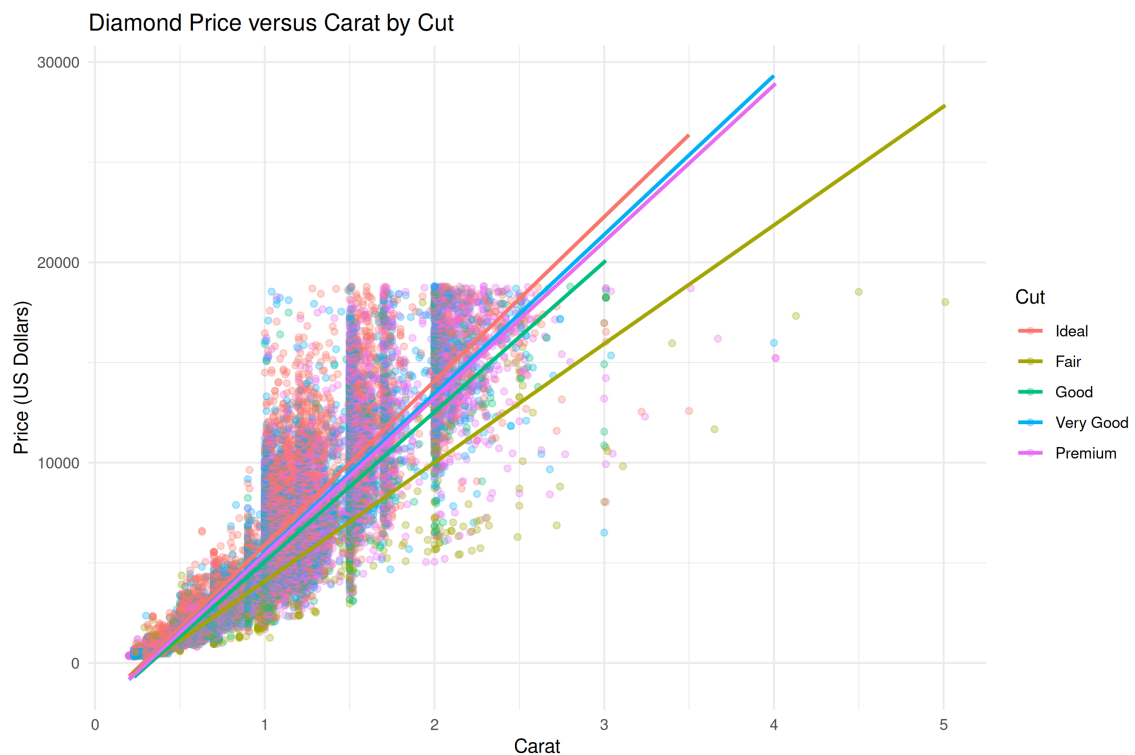


Figure 1: Diamond Price versus Carat by Cut

Statistical Conclusion

The results show a strong positive relationship between carat and diamond price, meaning that diamonds with greater carat weight generally have higher prices. Cut also plays a role in predicted diamond price, as shown by the dummy-variable coefficients relative to the Ideal reference category. The dummy variables represent the difference in predicted price between each cut category and the Ideal category while holding carat constant. The additive model assumes that the effect of carat on price is the same across all cut categories. However, the interaction test is statistically significant, providing evidence that the effect of carat on price differs among the cut categories. Overall, the regression analysis suggests that both carat and cut are important variables associated with diamond pricing.

Appendix. Complete R Commands Used

```
1 # Load the ggplot2 package
2 library(ggplot2)
3
4 # Load the diamonds dataset
5 data(diamonds)
6
7 # View the first few observations
8 head(diamonds)
9
10 # Check the structure of the dataset
11 str(diamonds)
12
13 # Get summary statistics
14 summary(diamonds)
15
16 # Number of observations
17 nrow(diamonds)
18
19 # Number of variables
20 ncol(diamonds)
21
22 # Convert cut to an unordered factor
23 diamonds$cut <- factor(diamonds$cut, ordered = FALSE)
24
25 # Set Ideal as the reference category
26 diamonds$cut <- relevel(diamonds$cut, ref = "Ideal")
27
28 # Check the categories
29 levels(diamonds$cut)
30
31
32 # PART B - CONSTRUCTING DUMMY VARIABLES
33
34 diamonds$Fair <- ifelse(diamonds$cut == "Fair", 1, 0)
35 diamonds$Good <- ifelse(diamonds$cut == "Good", 1, 0)
36 diamonds$Very_Good <- ifelse(diamonds$cut == "Very Good", 1, 0)
37 diamonds$Premium <- ifelse(diamonds$cut == "Premium", 1, 0)
38
39 head(
```

```

40 diamonds[, c(
41   "cut",
42   "Fair",
43   "Good",
44   "Very_Good",
45   "Premium"
46 )]
47 )
48
49
50 # PART C - ADDITIVE DUMMY-VARIABLE REGRESSION
51
52 model1 <- lm(price ~ carat + cut, data = diamonds)
53
54 summary(model1)
55 coef(model1)
56
57 cat("\n")
58 cat("ADDITIVE REGRESSION EQUATION\n")
59 cat("\n")
60
61 cat(
62   "Price = ",
63   round(coef(model1)[1], 3),
64   " + ",
65   round(coef(model1)[2], 3),
66   "(carat)",
67   " + cut effects\n"
68 )
69
70 cat("\nReference category: Ideal\n")
71
72 cat("\n")
73 cat(
74   "R-squared =",
75   round(summary(model1)$r.squared, 3),
76   "\n"
77 )
78
79 cat(
80   "Percentage of variation explained =",

```

```

81   round(summary(model1)$r.squared * 100, 3),
82   "%\n"
83 )
84
85
86 # PART D - INTERACTION BETWEEN CARAT AND CUT
87
88 model2 <- lm(price ~ carat * cut, data = diamonds)
89
90 summary(model2)
91
92 interaction_coefficients <- coef(summary(model2))
93
94 interaction_coefficients[
95   grep(
96     "carat:cut",
97     rownames(interaction_coefficients)
98   ),
99   ,
100   drop = FALSE
101 ]
102
103
104 # Hypotheses
105
106 cat("\n")
107 cat("H0: There is no interaction between carat and cut.\n")
108 cat(
109   "The effect of carat on price is the same for all cuts.\n"
110 )
111
112 cat("\n")
113 cat("H1: There is an interaction between carat and cut.\n")
114 cat(
115   "The effect of carat on price differs across cuts.\n"
116 )
117
118
119 # Incremental F-test
120
121 anova_result <- anova(model1, model2)

```

```

122
123 print(anova_result)
124
125 F_value <- anova_result$F[2]
126
127 p_value <- anova_result$`Pr(>F)`[2]
128
129 cat("\n")
130 cat(
131   "F-statistic =",
132   round(F_value, 3),
133   "\n"
134 )
135
136 cat(
137   "p-value =",
138   format.pval(p_value, digits = 4),
139   "\n"
140 )
141
142 alpha <- 0.05
143
144 if (p_value < alpha) {
145
146   cat("Decision: Reject H0.\n")
147
148   cat(
149     "There is statistically significant evidence of an\n",
150     "interaction between carat and cut.\n"
151   )
152
153 } else {
154
155   cat("Decision: Fail to reject H0.\n")
156
157   cat(
158     "There is not enough evidence of an interaction\n",
159     "between carat and cut.\n"
160   )
161 }
162

```

```

163
164 # PART E - VISUALIZATION
165
166 diamond_plot <- ggplot(
167   diamonds,
168   aes(
169     x = carat,
170     y = price,
171     color = cut
172   )
173 ) +
174   geom_point(alpha = 0.3) +
175   geom_smooth(
176     method = "lm",
177     se = FALSE
178   ) +
179   labs(
180     title = "Diamond Price versus Carat by Cut",
181     x = "Carat",
182     y = "Price (US Dollars)",
183     color = "Cut"
184   ) +
185   theme_minimal()
186
187 print(diamond_plot)
188
189 if (!dir.exists("figures")) {
190   dir.create("figures")
191 }
192
193 ggsave(
194   filename = "figures/diamond_price_by_cut.png",
195   plot = diamond_plot,
196   width = 9,
197   height = 6,
198   dpi = 300
199 )

```